



MTECH AAI COHORT-1

Dissertation Project – Phase II

Evaluation-I

Presented by: Sanjeev Kumar Pandey

Enrolment No: MT23AAI001

Project Guide: Dr. Amit Agrawal

Human ID & Activity Prediction using Wi-Fi Signals

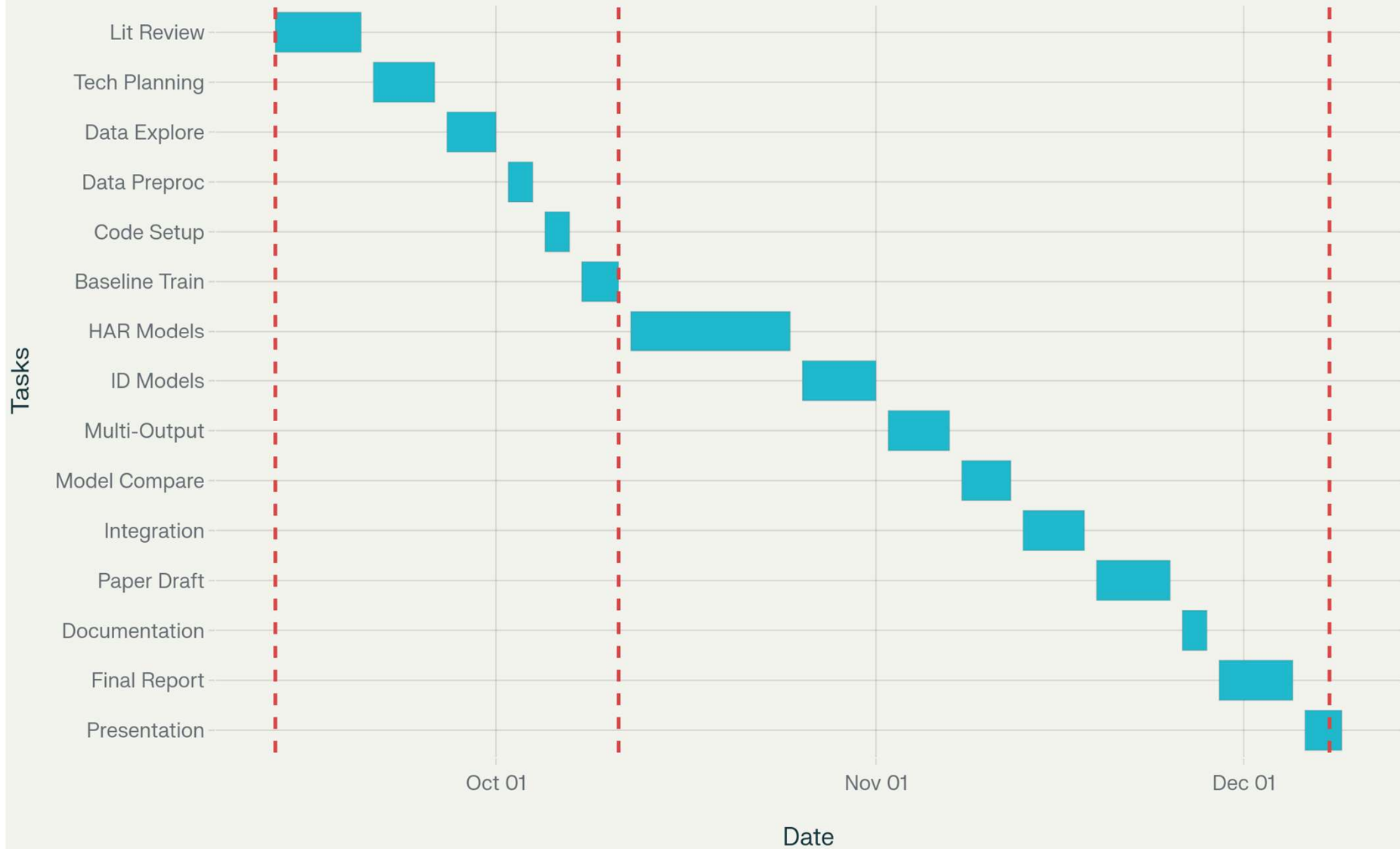
Leveraging Wi-Fi Channel State
Information (CSI) and RSSI

Sanjeev Kumar Pandey | 10-Sep-2025

Project Objectives

- **Dataset Aim:** Wi-Fi-based Human Activity Recognition (HAR) in indoor environments (LOS & NLOS).
- **Objective 1:** Detect the activity action type from Wi-Fi signals.
- **Objective 2:** Detect the ID of the person performing the activity.

3-Month Dissertation Project Timeline



Planning overview

- **Literature Review & Planning:** Start with reading key papers, setting objectives, and technical planning.
- **Data Understanding & Preprocessing:** Explore dataset, feature extraction, and handle CSI/RSSI preprocessing.
- **Initial Coding & Baseline Model (Eval 1):** Set up coding, implement baseline models, run initial training and evaluation.
- **Advanced Model Training & Experimentation (Eval 2):** Design and train advanced models (HAR, Human ID, multi-output), experiment, and tune.
- **Analysis & Integration:** Compare results, integrate best-performing models, and conduct error analysis.
- **Research Paper Drafting & Documentation:** Start documentation and research paper draft, plus slides for presentation.
- **Final Report & Presentation (Eval 3):** Polish final report, prepare for oral defense, and respond to committee feedback.

Dataset Overview

- **Source Paper:** 'A dataset for Wi-Fi-based human activity recognition in LOS and NLOS indoor environments'.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7704290/pdf/main.pdf>
- **Authors:** Baha' A Alsaify et al., Jordan University of Science and Technology, German Jordanian University.
- **Publication Date:** November 18, 2020.
- **Purpose:** Benchmark dataset for HAR research across LOS & NLOS settings.

Key Features of Dataset

- **Participants:** 30 subjects (28 male, 2 female), age $\sim 22.7 \pm 2.95$ years.
- **Activities:** 12 unique actions across 5 experiment classes.
- **Environments:** 2 LOS (lab, hallway) + 1 NLOS (wooden wall barrier).
- **Data Types:** Channel State Information (CSI) and RSSI values.
- **File Naming:** Ex_Sy_Cz_Ai_Tk.csv (e.g., E1_S04_C3_A7_T17.csv).

Feasibility of Human ID Prediction

- Subject ID explicitly labeled in filenames (S1–S30).
- Biometric info (age, weight, height) subtly affects signals.
- CSI & RSSI are sensitive to individual movement signatures.
- Rich dataset (3000 trials) suitable for Human ID modeling.

Use Case: Security & Monitoring

- Traditional biometrics (faces, fingerprints) can be spoofed & require active participation.
- Wi-Fi-based ID is passive, non-intrusive, and works in NLOS environments.
- **Applications:** High-security zones (ISRO, NASA, Parliament House), covert surveillance, medical alerts.

Target Variables

- **HAR Activity Recognition:** Classify 12 unique activities (A1–A12).
- **Human ID:** Identify 1 of 30 subjects (S1–S30).
- **Input Data:** CSI & RSSI values sensitive to presence & movement.

Model Development Approaches

- **Approach 1: Two Separate Models**
 - HAR Activity Classifier (12 classes).
 - Human ID Classifier (30 subjects).
- **Approach 2: One Multi-Output Model**
 - Single model with shared features.
 - Two output heads: Activity & ID.

Conclusion & Next Steps

- Dataset provides rich labeled Wi-Fi signal data (CSI, RSSI).
- Suitable for both HAR and Human ID tasks.
- Choice: Separate models (simplicity) vs. Multi-output model (efficiency).
- Next Step: Implement and evaluate both approaches.



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Dissertation Project – Phase II

Evaluation-II

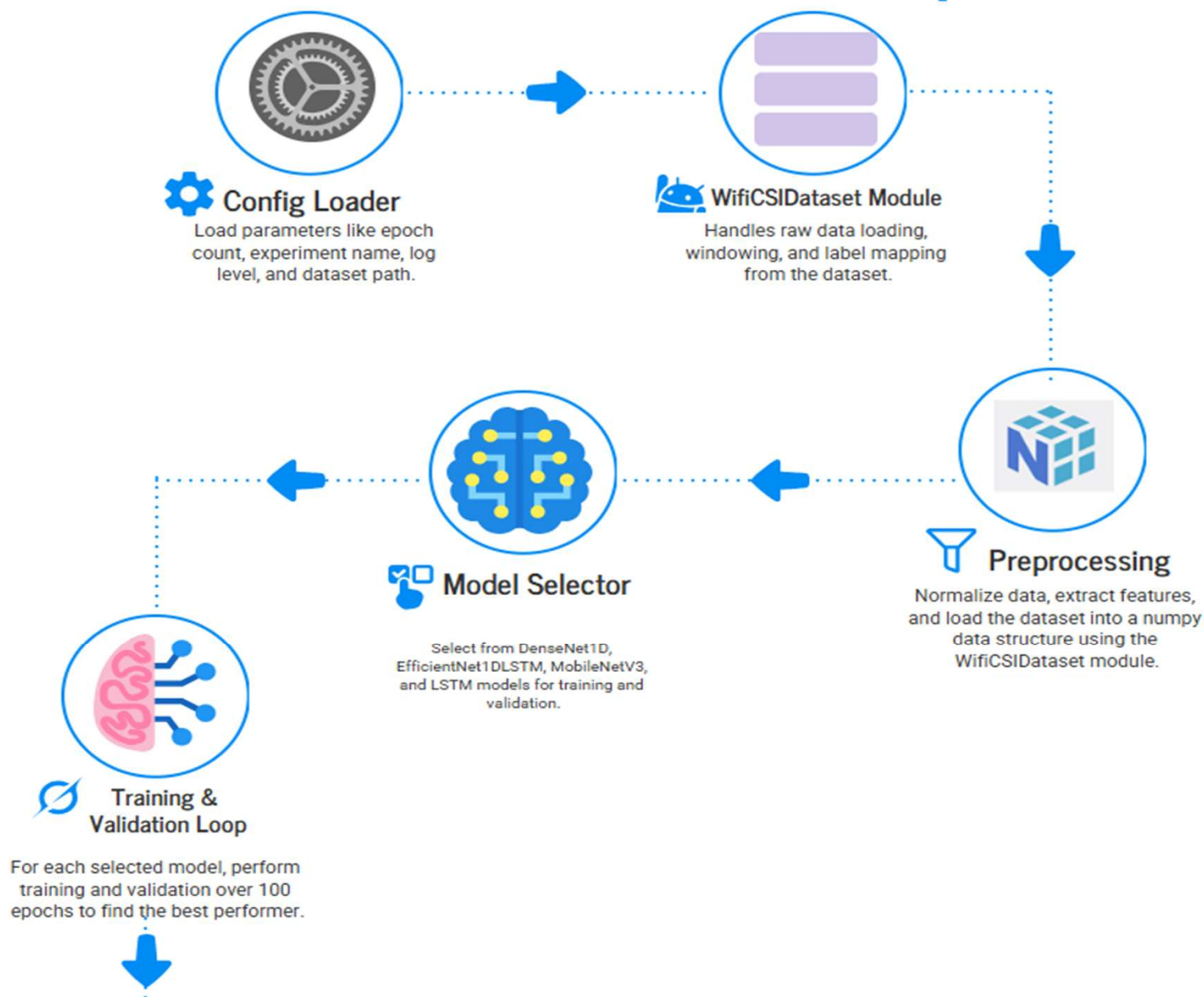
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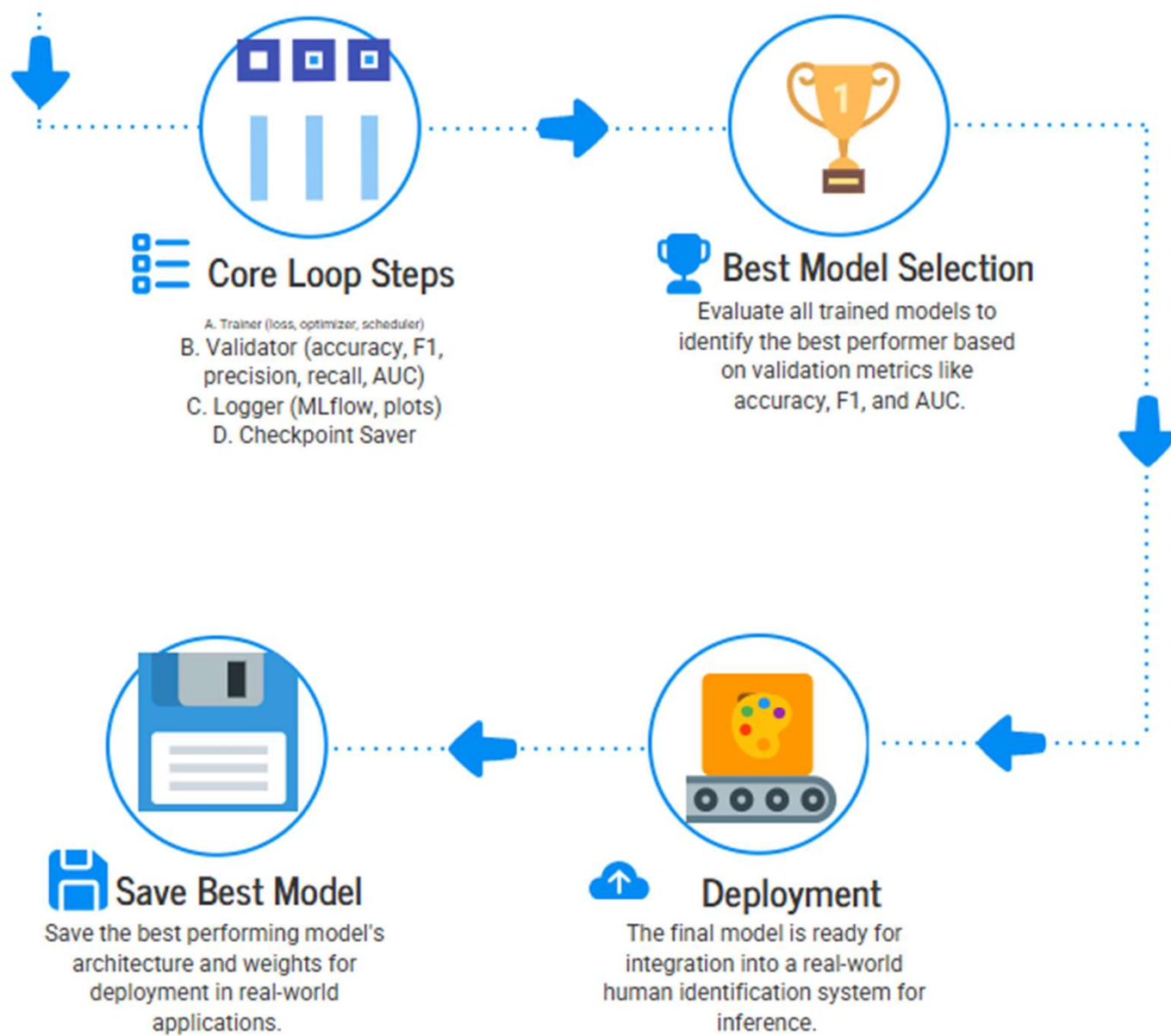
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Date: 11-Oct-2025

WiFi CSI-based Human ID Pipeline





Project Objective

The project performs a comparative study across four distinct DL architectures, all designed to process both the time-series CSI sequence and the auxiliary metadata sequence simultaneously.

Model Name	CSI Processing Core	Metadata Processing Core	Key Features
CSILSTMNet	Standard LSTM	Standard LSTM	Uses the last hidden state from two parallel LSTMs (one for CSI, one for Meta).
DenseNet1D	Dense Block 1D CNN	Bidirectional LSTM	Uses 1D CNN blocks (DenseBlock1D) followed by global pooling for CSI, combined with a Bi-LSTM for metadata.
EfficientNet1DLSTM	MBConv1D Blocks	Bidirectional LSTM	Leverages Mobile Inverted Bottleneck (MBConv) blocks (stem, mbconv1-3) designed for efficiency, combined with a Bi-LSTM.
MobileNetV3_1D_LSTM	MobileNetV3 Block 1D	Bidirectional LSTM	Incorporates specialized blocks (including depthwise separable convolutions and Squeeze-and-Excitation (SE) layers) optimized for mobile/efficient computing, combined with a Bi-LSTM.

MLflow Integration

- **Feature Focus:** Seamless integration with **MLflow** for experiment tracking and performance comparison.
- **What It Does:**
 - Automatically logs **training runs** for all four models — DenseNet1D, EfficientNet1DLSTM, MobileNetV3_1D_LSTM, and LSTM.
 - Captures **hyperparameters, metrics (Accuracy, F1, Precision, Recall, AUC)**, and **loss curves** per epoch.
- **Why It Matters:**
 - Enables **side-by-side model comparison** through MLflow dashboards and visual graphs.
 - Ensures **reproducibility** and transparency of experiments.
 - Simplifies identification of the **best-performing model** over 100 epochs.
- **Outcome:** Data-driven insights for model selection and performance optimization.

CSI Data

1 CSI Data Overview

Convert raw WiFi Channel State Information (CSI) signals into numerical features suitable for model training.

- **Nature of CSI:** Each CSI sample is a **complex number** representing the amplitude and phase response of each subcarrier in a WiFi link.

$$h_i(t) = a_i(t)e^{j\phi_i(t)} = \text{Re}(h_i(t)) + j \text{Im}(h_i(t))$$

- Here:
 - $a_i(t)$:amplitude (signal strength)
 - $\phi_i(t)$:phase shift caused by environment/human presence
 - i : subcarrier index, t : time index

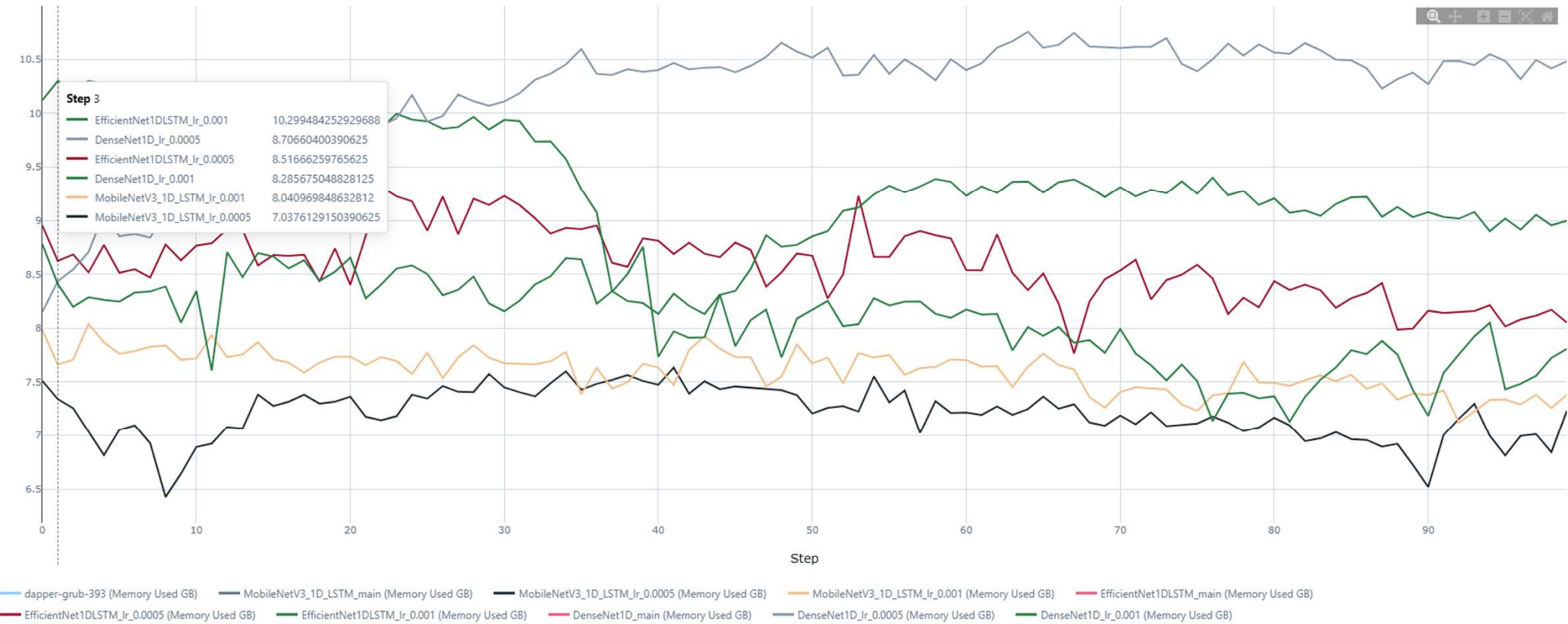
2 Feature Computation

- **Amplitude extraction:** converts complex CSI to real-valued form:

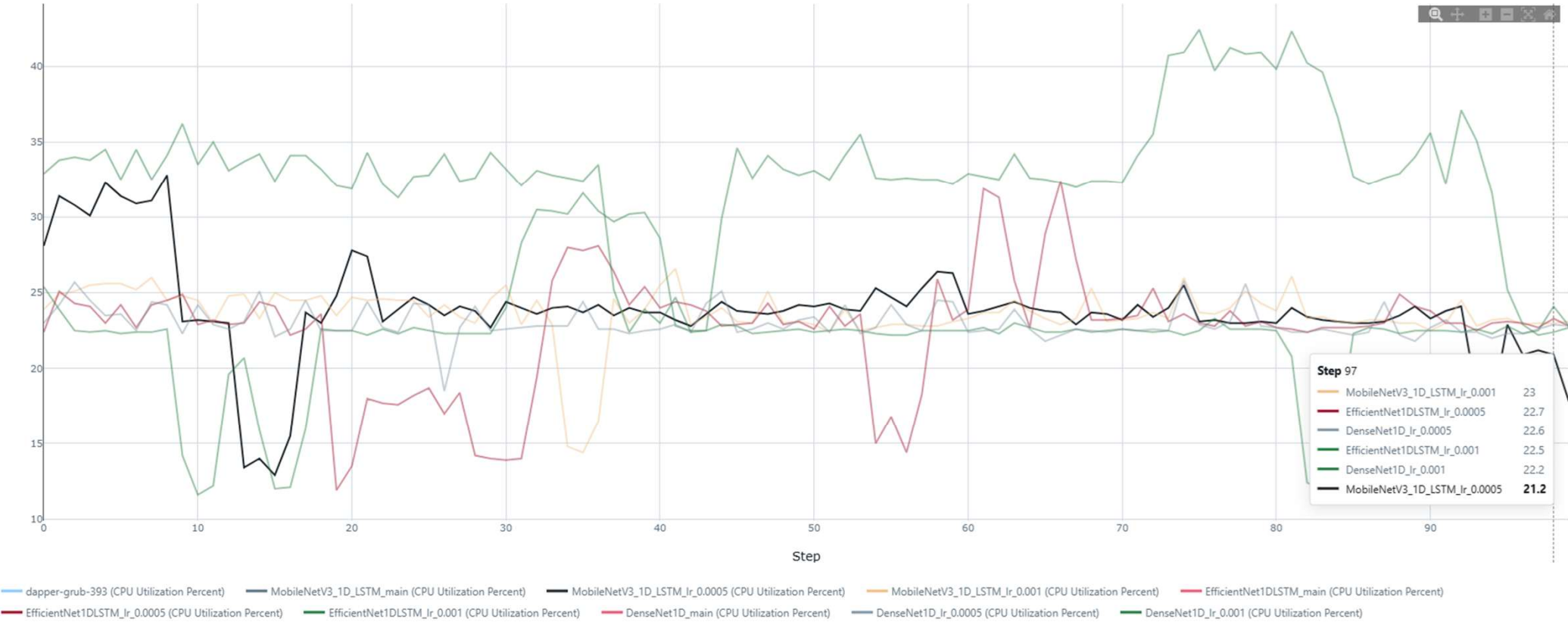
$$|h_i(t)| = \sqrt{(\text{Re}(h_i(t)))^2 + (\text{Im}(h_i(t)))^2}$$

```
115 meta_row = [float(row[c]) for c in meta_cols]
116 # CSI as magnitudes
117 csi_row = [abs(self.parse_complex(row[c])) for c in csi_cols]
118 x_meta.append(meta_row)
119 x_csi.append(csi_row)
```

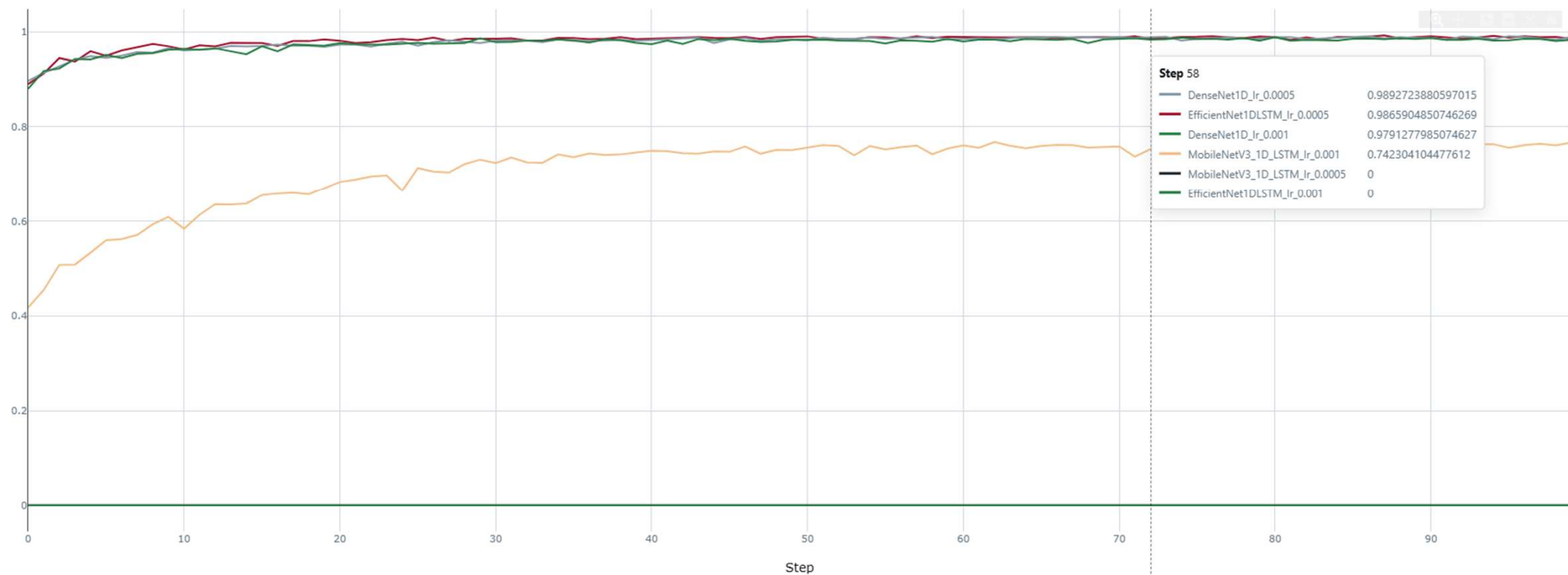
Memory Used GB



CPU Utilization Percent

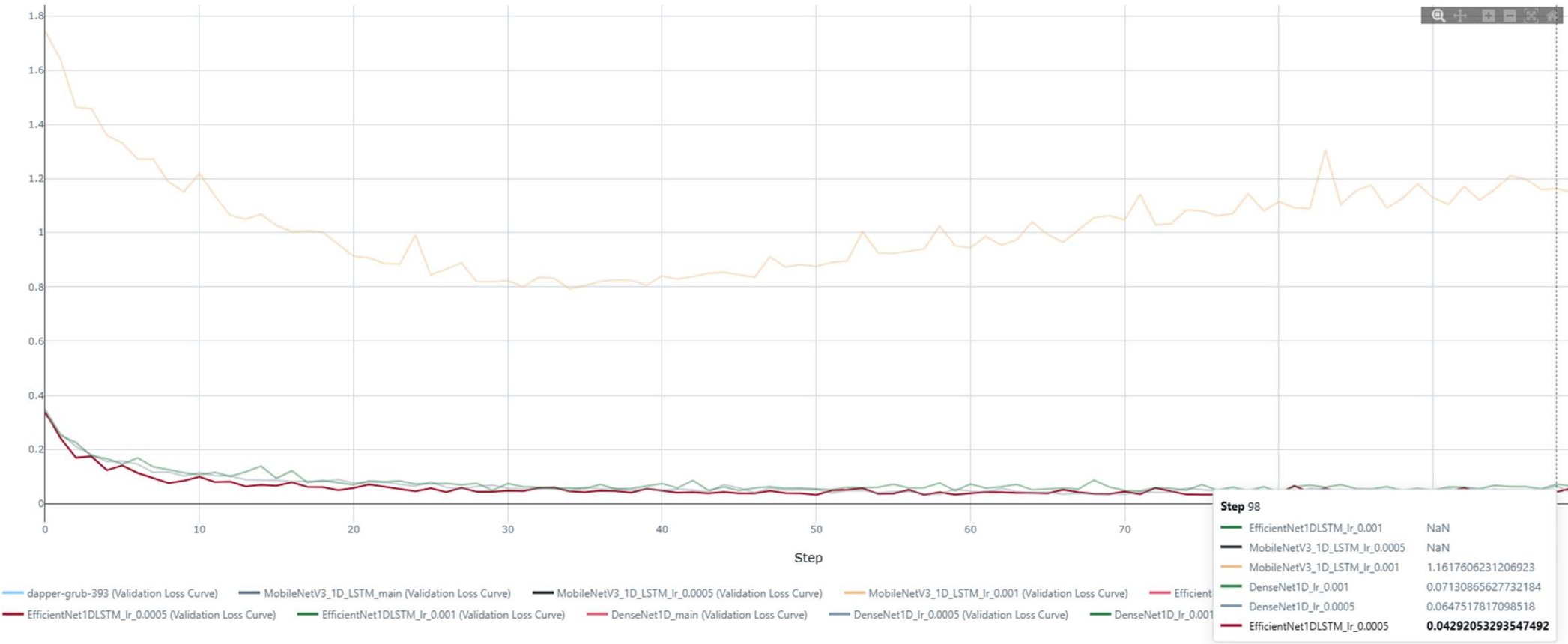


Validation Accuracy Curve



dapper-grub-393 (Validation Accuracy Curve)
 MobileNetV3_1D_LSTM_main (Validation Accuracy Curve)
 MobileNetV3_1D_LSTM_Ir_0.0005 (Validation Accuracy Curve)
 MobileNetV3_1D_LSTM_Ir_0.001 (Validation Accuracy Curve)
 EfficientNet1DLSTM_main (Validation Accuracy Curve)
 EfficientNet1DLSTM_Ir_0.0005 (Validation Accuracy Curve)
 EfficientNet1DLSTM_Ir_0.001 (Validation Accuracy Curve)
 DenseNet1D_main (Validation Accuracy Curve)
 DenseNet1D_Ir_0.0005 (Validation Accuracy Curve)
 DenseNet1D_Ir_0.001 (Validation Accuracy Curve)

Validation Loss Curve



Best Model

Model Performance Comparison

- Objective: Evaluate four DL models on experiment data using two learning rates (0.001, 0.0005)

- Top Performing Models:

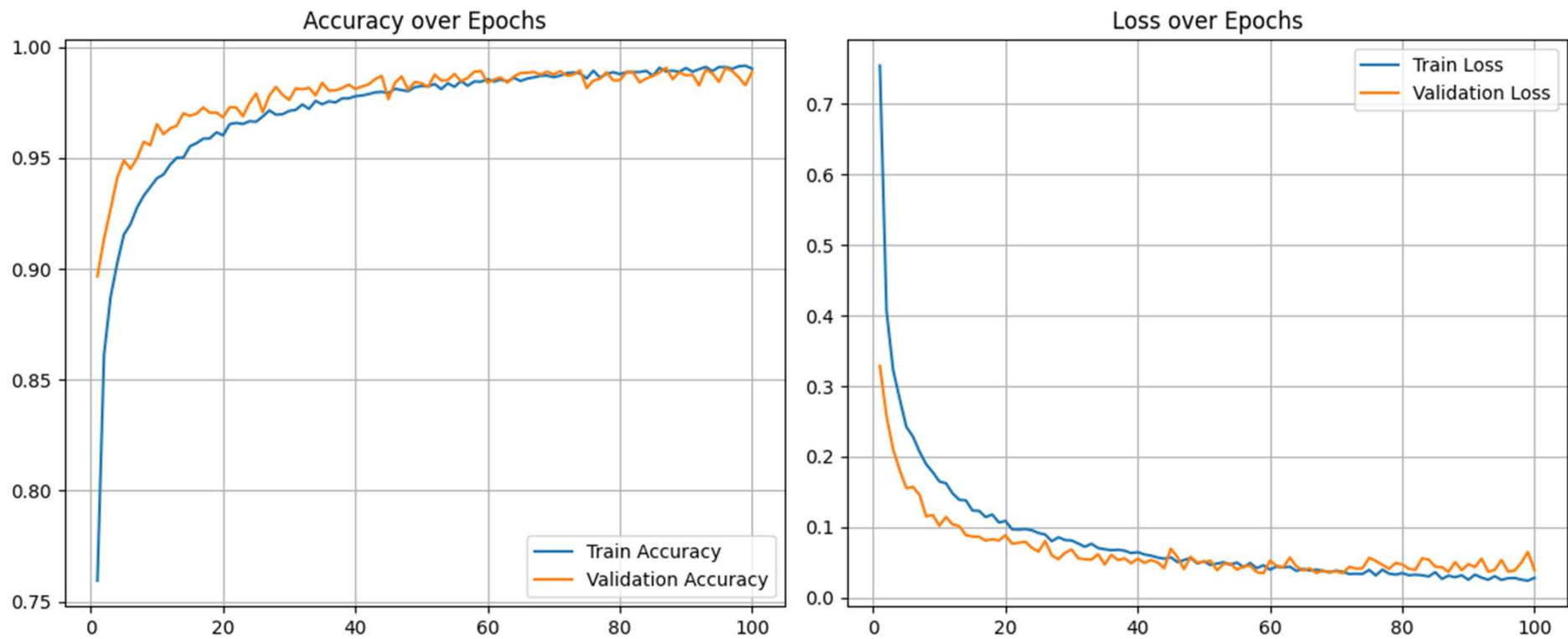
- DenseNet1D (lr=0.0005): Best overall metrics
- EfficientNet1DLSTM (lr=0.0005): High, consistent metrics
- CSILSTMNet (lr=0.0005, 0.001): Slightly below leaders
- MobileNetV3_1D_LSTM (lr=0.001): Lower but strong metrics

Model	Learning Rate	Best Validation Accuracy	Best Validation Precision	Best Validation Recall	Best Validation F1 Score
DenseNet1D	0.0005	0.999938	0.999938	0.999938	0.999938
EfficientNet1DLSTM	0.0005	0.99993	0.99993	0.99993	0.99993
DenseNet1D	0.001	0.999904	0.999904	0.999904	0.999904
CSILSTMNet	0.0005	0.999826	0.999826	0.999826	0.999826
CSILSTMNet	0.001	0.999814	0.999814	0.999814	0.999814
MobileNetV3_1D_LSTM	0.001	0.988529	0.988529	0.988529	0.988529

Insights & Conclusions

- **Learning Rate Tuning:** Lower learning rate (0.0005) consistently outperforms 0.001 for all models
- **Metric Trends:** Metrics (accuracy, precision, recall, F1) closely match for each model configuration, indicating balanced performance
- **Best Model:** DenseNet1D with $\text{lr}=0.0005$ is the optimal choice for this experimental dataset
- **Generalization:** High and consistent validation scores suggest good model generalization and minimal overfitting issues across the experiments

DenseNet1D with $\text{lr}=0.0005$ Plot



DenseNet1D with lr=0.0005 Plot

	precision	recall	f1-score	Classification Report support
1	0.99	1.00	1.00	277
2	0.99	0.92	0.96	283
3	0.99	1.00	1.00	292
4	1.00	1.00	1.00	288
5	0.94	0.99	0.96	272
6	0.98	0.99	0.99	295
7	1.00	1.00	1.00	290
8	0.99	1.00	0.99	267
9	1.00	0.99	1.00	267
10	0.99	0.99	0.99	272
11	1.00	0.99	1.00	270
12	1.00	1.00	1.00	272
13	0.99	0.95	0.97	332
14	1.00	1.00	1.00	301
15	1.00	0.99	0.99	253
16	0.99	1.00	1.00	307
17	0.99	0.98	0.98	290
18	0.98	1.00	0.99	306
19	0.95	0.99	0.97	282
20	0.99	1.00	0.99	285
21	0.97	1.00	0.98	289
22	0.99	0.97	0.98	316
23	0.98	1.00	0.99	288
24	1.00	1.00	1.00	272
25	1.00	0.98	0.99	308
26	1.00	1.00	1.00	263
27	1.00	1.00	1.00	287
28	1.00	1.00	1.00	293
29	1.00	0.97	0.98	294
30	0.97	0.98	0.98	265
accuracy			0.99	8576
macro avg	0.99	0.99	0.99	8576
weighted avg	0.99	0.99	0.99	8576

THANK YOU